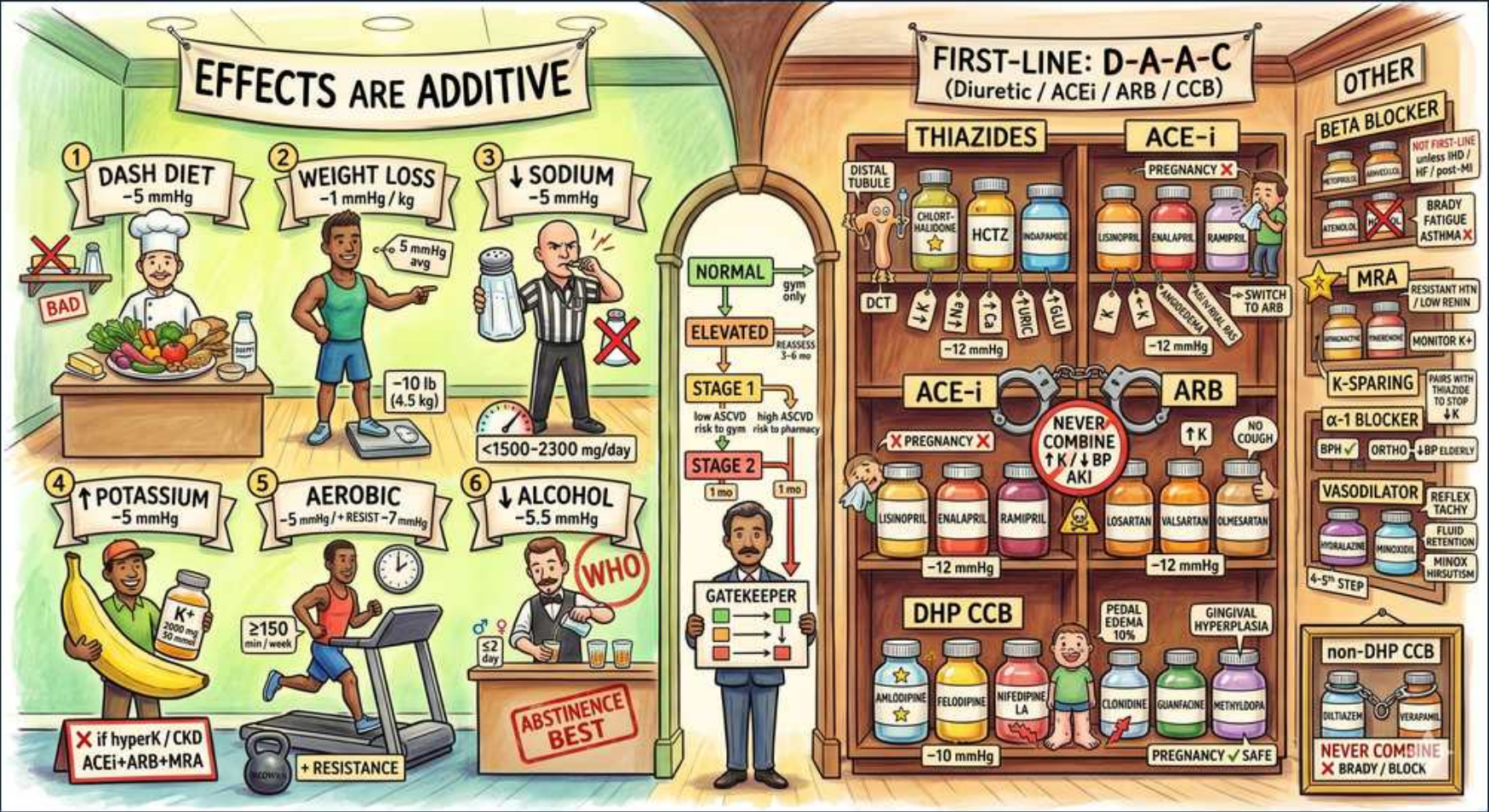


Hypertension — Antihypertensive Classes & Lifestyle



1. First-line: D-A-A-C

WHAT YOU SEE

The big gold FIRST-LINE: D-A-A-C banner across the top of the pharmacy wall — the gold header marks these four shelves (Diuretic / ACEi / ARB / CCB) as the go-to starting classes.

WHAT IT ENCODES

Four first-line classes for uncomplicated HTN: thiazide Diuretic, ACE inhibitor, ARB, and CCB (the 'D-A-A-C' shelf). Any one can start uncomplicated HTN; choice is individualized (e.g., ACEi/ARB in diabetes/CKD, thiazide/CCB often preferred in Black patients, CCB/thiazide in isolated systolic HTN of elderly). Stage 2 HTN typically begins with TWO agents from different classes.

BOARD TRAP

WRONG: choosing a beta-blocker as first-line for uncomplicated essential HTN. Discriminator — beta-blockers are NOT first-line absent a compelling indication (post-MI, HFrEF, AFib, angina); they are less effective at preventing stroke than the D-A-A-C classes.

2. Thiazides

WHAT YOU SEE

THIAZIDES shelf, HCTZ / chlorthalidone bottles, distal tubule

WHAT IT ENCODES

Thiazides (hydrochlorothiazide 12.5-50 mg, chlorthalidone 12.5-25 mg, indapamide) block the Na-Cl cotransporter (NCC) in the distal convoluted tubule. Chlorthalidone is longer-acting and more potent than HCTZ. Metabolic effects: hypokalemia, hypoNATREMIA, metabolic alkalosis, hyperuricemia (gout), hyperGLYCEMIA, hyperlipidemia, and hyperCALCEMIA. Remember the calcium direction: thiazides RAISE calcium (used in calcium-stone prevention and hypercalciuria) whereas loops LOWER it.

BOARD TRAP

WRONG: continuing/starting a thiazide in a patient who develops gout or in one with recurrent calcium kidney stones being worked up — and confusing the calcium direction. Discriminator — thiazides cause hyperuricemia (precipitate gout) and HYPERcalcemia (loops cause hypocalcemia); if a stem shows low Na + low K + high Ca + high uric acid, think thiazide, not a loop diuretic.

3. ACE inhibitors

WHAT YOU SEE

ACE-i shelf, -pril bottles, pregnancy X, switch to ARB if cough

WHAT IT ENCODES

ACE inhibitors ('-pril': lisinopril 10-40 mg, enalapril, ramipril) block conversion of angiotensin I to II and reduce bradykinin breakdown. Effects: lower BP, reduce proteinuria, renoprotective in diabetic nephropathy/CKD, and improve survival in HFrEF and post-MI. Side effects: dry COUGH and ANGIOEDEMA (both bradykinin-mediated), HYPERKALEMIA, acute kidney injury in bilateral RAS, and teratogenicity. A mild creatinine rise up to ~30% is acceptable.

BOARD TRAP

WRONG: switching an ACEi-induced ANGIOEDEMA patient to an ARB. Discriminator — for the dry COUGH, switching to an ARB is correct (ARBs don't raise bradykinin); but after ANGIOEDEMA, ARBs carry cross-reactivity risk and the safer move is a different class entirely (e.g., CCB).

4. ARBs

WHAT YOU SEE

ARB shelf, -sartan bottles

WHAT IT ENCODES

ARBs ('-sartan': losartan 50-100 mg, valsartan, candesartan) block the angiotensin II type-1 receptor. Same hemodynamic and renoprotective benefits as ACEi but WITHOUT cough or significant bradykinin-mediated angioedema. They still raise potassium, can cause AKI in bilateral RAS, and are TERATOGENIC. First choice when an ACEi causes intolerable cough.

BOARD TRAP

WRONG: assuming ARBs are safe in pregnancy because they 'aren't ACE inhibitors.' Discriminator — ARBs share the same fetal renal/skull toxicity and oligohydramnios as ACEi and are contraindicated in pregnancy; the difference from ACEi is only the cough/angioedema profile, not teratogenicity.

5. NEVER combine ACEi+ARB

WHAT YOU SEE

NEVER COMBINE warning between ACEi and ARB shelves

WHAT IT ENCODES

Do NOT combine an ACEi with an ARB. Dual RAAS blockade gives no added cardiovascular/renal benefit and significantly increases hyperkalemia, hypotension, and acute kidney injury (ONTARGET trial). The same caution applies to adding a direct renin inhibitor (aliskiren) to an ACEi/ARB, especially in diabetics.

BOARD TRAP

WRONG: adding an ARB to maximize renoprotection in a diabetic already on an ACEi with persistent proteinuria. Discriminator — dual RAAS blockade harms (AKI, hyperkalemia) without outcome benefit; instead optimize the single agent and add a different class (e.g., MRA, SGLT2 inhibitor) or a diuretic.

6. DHP CCB

WHAT YOU SEE

DHP CCB shelf, -dipine bottles, pedal edema

WHAT IT ENCODES

Dihydropyridine CCBs ('-dipine': amlodipine 5-10 mg, nifedipine, felodipine) block vascular L-type calcium channels -> potent arterial VASODILATION with little effect on the heart's conduction. Side effects: dose-dependent peripheral/PEDAL EDEMA (from precapillary dilation, not fluid overload), flushing, headache, reflex tachycardia, and gingival hyperplasia. Good in isolated systolic HTN of the elderly and in Black patients.

BOARD TRAP

WRONG: treating amlodipine-induced ankle edema with a loop diuretic. Discriminator — DHP edema is from arteriolar dilation/capillary pressure, NOT volume overload, so diuretics don't help; the fix is dose reduction or adding an ACEi/ARB (venodilation balances the precapillary dilation).

7. Non-DHP CCB

WHAT YOU SEE

non-DHP CCB shelf, verapamil/diltiazem, bradycardia/block

WHAT IT ENCODES

Non-dihydropyridine CCBs — verapamil (most cardioselective, also constipation) and diltiazem — block cardiac L-type calcium channels: they slow the SA/AV node (negative chronotrope/dromotrope) and reduce contractility (negative inotrope). Useful for rate control in AFib/SVT and for angina. Side effects: bradycardia, AV block, worsening HFrEF, and constipation (verapamil).

BOARD TRAP

WRONG: adding verapamil or diltiazem to a patient already on a beta-blocker. Discriminator — both suppress the AV node and contractility; the combination risks profound bradycardia, complete heart block, and decompensated heart failure. Also avoid non-DHP CCBs in HFrEF.

8. Beta-blockers

WHAT YOU SEE

The BETA BLOCKER shelf parked over in the OTHER section with a bradycardia heart icon and a 'not first-line' tag — being off to the side, not on the main D-A-A-C wall, is the visual cue it needs a compelling indication.

WHAT IT ENCODES

Beta-blockers (metoprolol, carvedilol, bisoprolol, atenolol) reduce HR, contractility, and renin release. NOT first-line for uncomplicated HTN, but indicated with compelling reasons: post-MI, HFrEF (carvedilol/metoprolol succinate/bisoprolol), rate control in AFib, angina, and certain tachyarrhythmias. Side effects: bradycardia, fatigue, masking of hypoglycemia symptoms, bronchospasm (non-selective), and rebound HTN/tachycardia if stopped abruptly.

BOARD TRAP

WRONG: abruptly discontinuing a beta-blocker in a patient with coronary disease. Discriminator — abrupt withdrawal causes receptor up-regulation and rebound tachycardia/HTN, precipitating angina, MI, or arrhythmia; beta-blockers must be tapered.

9. Spironolactone / MRA

WHAT YOU SEE

The MRA shelf marked K-SPARING with a 'monitor K' warning — flagged as the add-on for RESISTANT HTN.

WHAT IT ENCODES

Mineralocorticoid receptor antagonists — spironolactone 25-50 mg (eplerenone is more selective) — block aldosterone at the distal nephron/collecting duct: potassium-sparing diuresis. They are the evidence-based ADD-ON of choice for RESISTANT HTN (uncontrolled on a thiazide + ACEi/ARB + CCB, per PATHWAY-2) and treat primary aldosteronism and HFrEF. Spironolactone also causes GYNECOMASTIA, breast tenderness, and menstrual irregularity (anti-androgen effect); eplerenone avoids these.

BOARD TRAP

WRONG: continuing spironolactone in a man who develops painful gynecomastia, or stacking it on an ACEi/ARB without checking K+. Discriminator — switch to eplerenone for the gynecomastia (no anti-androgen effect); and combining an MRA with ACEi/ARB markedly raises hyperkalemia risk, so potassium and creatinine must be monitored closely.

10. Gatekeeper / staging

WHAT YOU SEE

Central pillar Normal/Elevated/Stage1/Stage2 gatekeeper figure

WHAT IT ENCODES

ACC/AHA staging gates therapy: Normal <120/80; Elevated 120-129/<80; Stage 1 130-139 or 80-89; Stage 2 ≥140 or ≥90. Stage 1 with <10% 10-year ASCVD risk → lifestyle first; Stage 1 with high risk (ASCVD, diabetes, CKD, ≥10% risk) or any Stage 2 → start drug therapy. Stage 2 usually starts with TWO drugs from different classes. General target <130/80.

BOARD TRAP

WRONG: starting two drugs for every patient labeled 'hypertensive,' including low-risk Stage 1. Discriminator — initiate two-drug therapy for Stage 2 (or markedly above goal); low-risk Stage 1 gets lifestyle first, then a single agent if needed. Threshold and intensity scale with stage AND CV risk.

11. Lifestyle is additive

WHAT YOU SEE

EFFECTS ARE ADDITIVE banner over lifestyle panels

WHAT IT ENCODES

Lifestyle interventions lower BP independently and ADDITIVELY — stacking DASH, sodium restriction, weight loss, exercise, potassium, and alcohol moderation can total 10-15+ mmHg, rivaling a drug. They are foundational at every stage and continue alongside medication; they also improve drug efficacy and CV risk profile.

BOARD TRAP

WRONG: abandoning lifestyle counseling once medication is started. Discriminator — lifestyle effects are additive to drug effects and reduce the number/dose of drugs needed; they remain part of the plan after pharmacotherapy begins.

12. DASH diet

WHAT YOU SEE

DASH DIET chef, ~11 mmHg

WHAT IT ENCODES

The DASH diet (fruits, vegetables, low-fat dairy, whole grains; limited saturated fat/red meat/sweets) gives the LARGEST single lifestyle BP drop, up to ~11 mmHg SBP in hypertensives, and is even larger when combined with sodium restriction. It also raises dietary potassium and fiber.

BOARD TRAP

WRONG: equating a generic 'low-fat diet' with DASH. Discriminator — DASH's BP benefit comes from its specific high-produce, high-potassium, high-dairy, low-sodium pattern; a nonspecific low-fat diet does not reproduce the ~11 mmHg effect.

13. Sodium & weight

WHAT YOU SEE

SODIUM <1500-2300mg/day; WEIGHT LOSS ~1mmHg/kg

WHAT IT ENCODES

Sodium restriction to ~1500-2300 mg/day lowers SBP ~5 mmHg (more in salt-sensitive: older, Black, CKD, diabetic patients), and weight loss lowers SBP ~1 mmHg per kilogram lost. Both are dose-dependent and additive with DASH. Most sodium is from processed/restaurant foods.

BOARD TRAP

WRONG: expecting little benefit from modest weight loss in an obese patient. Discriminator — BP falls ~1 mmHg per kg, so even a 5-10% body-weight loss yields a meaningful drop; benefit is proportional and does not require reaching ideal weight.

14. Potassium / aerobic / alcohol

WHAT YOU SEE

POTASSIUM ~-5; AEROBIC ~-5; ALCOHOL -5.5, abstinence best

WHAT IT ENCODES

Three more additive measures each lower SBP ~5 mmHg: increasing dietary potassium (~3500-5000 mg/day, food-based), aerobic exercise (≥ 150 min/week moderate intensity), and reducing alcohol (men ≤ 2 /day, women ≤ 1 /day; abstinence best). Potassium's effect is larger in Black adults.

BOARD TRAP

WRONG: recommending potassium supplementation/salt substitutes to a CKD patient or one on an ACEi/ARB/MRA. Discriminator — these patients can't excrete a potassium load and risk dangerous hyperkalemia; food-based potassium counseling is fine only when renal function and RAAS-blockade allow.

15. Loop diuretics (volume/CKD)

WHAT YOU SEE

The loop/distal tubule shelf reserved for CKD / volume overload — placed apart from the thiazide shelf to signal it is the diuretic to switch to when GFR is low.

WHAT IT ENCODES

NEW: Loop diuretics — furosemide, torsemide, bumetanide — block the Na-K-2Cl cotransporter in the thick ascending limb (most powerful natriuresis). In HTN they are not first-line but are PREFERRED over thiazides when GFR is low (eGFR < 30) or with volume overload/heart failure, because thiazides lose efficacy in advanced CKD. Side effects: hypokalemia, hypocalcemia (calcium WASTING), hypomagnesemia, metabolic alkalosis, ototoxicity, and hyperuricemia.

BOARD TRAP

WRONG: choosing a thiazide for HTN in a patient with advanced CKD (eGFR < 30). Discriminator — thiazides are largely ineffective at low GFR; switch to a loop diuretic. Also remember loops LOWER calcium (used for hypercalcemia) whereas thiazides RAISE it — opposite calcium effects.

16. Hydralazine / direct vasodilators

WHAT YOU SEE

The VASODILATOR shelf showing reflex-tachycardia hearts (hydralazine / minoxidil) — the racing heart pictures the reflex tachycardia these direct vasodilators trigger.

WHAT IT ENCODES

NEW: Direct arteriolar vasodilators. Hydralazine reduces afterload (used IV in pregnancy/hypertensive emergencies and in the hydralazine-isosorbide dinitrate combo for HFrEF in Black patients); it causes reflex tachycardia and DRUG-INDUCED LUPUS (anti-histone antibodies). Minoxidil is a potent K-channel-opener vasodilator reserved for refractory HTN; it causes marked fluid retention and HYPERTRICHOSIS (the basis for topical hair-loss use).

BOARD TRAP

WRONG: missing drug-induced lupus in a patient on hydralazine who presents with arthralgias, fever, and serositis. Discriminator — hydralazine (and procainamide, isoniazid) cause a lupus-like syndrome with anti-HISTONE antibodies that resolves on stopping the drug; it typically spares the CNS and kidneys unlike idiopathic SLE.

17. Alpha-1 blockers (-zosin)

WHAT YOU SEE

The alpha/'B1' blocker shelf tagged BPH plus a first-dose-hypotension warning — pairing prostate + falling-BP cues marks it as the HTN-with-BPH agent that causes first-dose orthostasis.

WHAT IT ENCODES

NEW: Alpha-1 blockers ('-zosin': doxazosin, terazosin, prazosin) relax vascular and prostatic smooth muscle. Not first-line for HTN (ALLHAT showed worse heart-failure outcomes vs chlorthalidone), but they are the antihypertensive of choice when HTN coexists with BPH (treat both). Classic side effect: FIRST-DOSE orthostatic hypotension/syncope — start low, dose at bedtime.

BOARD TRAP

WRONG: selecting an alpha-blocker as first-line monotherapy for uncomplicated HTN. Discriminator — ALLHAT demonstrated increased heart failure with doxazosin vs a thiazide; reserve alpha-blockers for the BPH-plus-HTN patient, and warn about first-dose orthostatic hypotension.